SCHRÖDINGER EVOLUTION AT THE $\alpha \to 1^-$ EDGE OF THE CAUSAL ACCUMULATION LAW: DERIVED UNITARITY, THE FREE GALILEAN LIMIT, AND CARRIER ENTANGLEMENT AS TIME DILATION

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ORCID: 0009-0000-0133-2291. Companion to *The Causal Accumulation Law: The Four Corners of α, Algebraic Closure, and the Standard Model* [1] and to *The $\alpha = 1$ Unitary Skeleton and the Emergence of Schrödinger Evolution in the δ-Kernel Limit of the CAL* [2], whose unitarity posit it replaces. Dated November 11, 2026.

Abstract. We impose the Causal Accumulation Law (CAL) as a field equation, $T = 0$, with a power-law memory under causal Weyl integration and the channel orientation of the parent framework carried by the memory channel. As the fractional order approaches $\alpha = 1$ from below, the field equation collapses to the Schrödinger equation. The kinetic term is not assumed. Keeping the full spatial memory kernel, it is the long-wavelength scaling limit in which the kernel's spatial width shrinks at fixed $\hbar^2/2m$, and no energy cap remains.

Unitarity is derived, not posited. For every $\alpha < 1$, the bounded solutions on the whole time line form a strongly continuous unitary group, because the channel phase cancels the phase of the Weyl symbol on one sign of frequency. This replaces the unitarity posit of the companion paper.

The limit is one-sided. The Weyl integral converges on oscillating carriers only for $\alpha < 1$, and the memory-generated coefficients change sign across $\alpha = 1$. Above $\alpha = 1$ the one-way edge equation has no causal, bounded continuation: its response to a source either grows at rate $\pi|1-\alpha|\,|E|/\hbar$ or precedes the source. This is a property of that equation, not of the framework's dynamics on $(1, 2]$, which the $\alpha = 2$ closure governs.

The kernel-blind skeleton at $\alpha = 1$ and the Schrödinger edge are two scalings of one limit. With the kernel's time unit held fixed, the memory rate vanishes and the past disconnects even from below. The edge exists only if the memory rate $(1 - \alpha)/\tau_c$ is held fixed. In that limit the kernel becomes a flat accumulator, not a δ.

The framework's two Markov limits are the forgetful (Dirac) limit and the faithful (full light-cone) limit [3]. A closed system cannot reach the faithful one. The edge reaches neither: it is Markov through the fundamental theorem of calculus.

- A Dirac, or any normalizable, memory kernel on the carrier has no stationary band at all. The plane-wave equation needs $|\hat{w}| = \alpha/(\varepsilon|\lambda|)$, while $|\hat{w}| \leq 1$, so every mode decays.
- As $\alpha \to 1^-$ at fixed memory rate, the Weyl kernel becomes the flat accumulator, whose inverse is ∂_t. It keeps the accumulated sum, not the individual contributions, so it is not the faithful limit. For $\alpha < 1$ a Markov description of the power-law memory needs a continuum of auxiliary modes. As $\alpha \to 1^-$ that continuum collapses onto a single mode, the state itself. The edge is therefore Markov without enlarging the state space and without keeping records.
- The power law reaches this limit because it is not normalizable, and because the phase of its symbol is the channel angle.

The edge is Galilean and non-interacting.

- The Lorentz signature belongs to the $\alpha = 2$ closure, so the edge has no light cone. This is consistent with the instantaneous spatial reach of the Schrödinger propagator.
- Its generator is built from one-way transports. It therefore gives the free Schrödinger form, at most in a fixed one-way background.
- The edge operator contains only single-carrier, one-way transport. A coupling between carriers would need its adjoint, the return leg, to keep the evolution unitary, and that reciprocal structure belongs to the $\alpha = 2$ closure. So the edge cannot create entanglement between carriers.

A single carrier $J \in M_2(\mathbb{C}) \cong \mathbb{C}^2 \otimes \mathbb{C}^2$ is itself a two-factor object, and $JJ^\dagger$ is the reduced state of its left factor. Left $SL(2, \mathbb{C})$ acts on that closure as the Lorentz group: rotations leave the entanglement between the two factors unchanged, and boosts change it.

- For a current of rapidity η, the concurrence is $1/\gamma = \operatorname{sech}\eta$, and the Schmidt weights are proportional to the Doppler factors $e^{\pm\eta}$.
- So a carrier's internal entanglement is the time dilation of its current: maximal at rest, and zero for a massless carrier.
- The trace readout $\mathrm{tr}(P_m JJ^\dagger)/\mathrm{tr}(JJ^\dagger)$ is unchanged under tensor composition, at any rank.

Numerical runs of the field equation, using Weyl integration only, confirm each statement.

Contents

1. Scope and claim 3
2. Definitions 4
3. Axioms and the field equation 5
4. Unconditional lemmas 5
5. Derived unitarity and the collapse 6
6. The two Markov limits, causal structure, interaction, and readout 9
6.1. The corner and the accumulator 10
6.2. A Dirac memory kernel: no stationary band 10
6.3. Causal structure: the edge is Galilean 11
6.4. The edge carries no interaction 11
6.5. Readout: one carrier, and composites 12
7. Numerical verification 14
8. Remarks on scope 17
Appendix A. Notation 18
Appendix B. Numerical methods 18
Appendix C. For reference: a light cone imposed by hand 18
References 19

1. Scope and claim

The companion paper [2] obtains the Schrödinger form at the edge of the (α, kernel) table adjoining $\alpha = 1$ from below, but it consumes an input: Posit 1 assumes a unitary flow $U(t) = e^{i\theta(t)G}$ and then defines $\hat{H} \equiv i\hbar \dot{U} U^{-1}$. That step establishes the form of the equation only after unitarity and a generator have been assumed. The present paper removes the posit. The accumulation law itself, read as a field equation, collapses to the Schrödinger equation, and unitarity, the one-sidedness of the limit, and the mass formula follow.

> *Impose $T = 0$ with a power-law memory under causal Weyl integration, and let the memory channel carry the orientation $e^{-i\sigma\alpha\pi/2}$ of Axiom 3. For every $\alpha < 1$, the bounded solutions on the whole time line evolve by a unitary group (Theorem 5.1). As $\alpha \to 1^-$ with the memory rate held fixed, they converge to Schrödinger evolution (Theorem 5.3). Above $\alpha = 1$ the same equation has no causal, bounded continuation (Theorem 5.8). The kinetic term is a scaling limit of the full spatial kernel (Proposition 5.4). A Dirac, or any normalizable, memory kernel on the carrier cannot do this, because it has no stationary band (Theorem 6.2). The edge is Markov through the fundamental theorem of calculus instead, without enlarging the state space (Proposition 6.5). The edge is Galilean and carries no interaction (§6.3–6.4), and its trace readout survives tensor composition (§6.5). The carrier's internal entanglement is the time dilation of its current (Proposition 6.8).*

All dynamics in this paper use Weyl integration over the whole past, which is the integral the accumulation law writes. Initial-value (Caputo) problems are not used. A Caputo problem is the Weyl derivative of a history frozen at its initial value for all earlier times. Such a frozen past has zero-frequency content, on which the Weyl integral of the field equation diverges, so a Caputo problem is not a solution of Axiom 3. Starting a solution is instead modelled by a source acting on a zero past (Proposition 5.6).

Bookkeeping. The inputs are:

- the field equation (Axiom 3), which replaces Posit 1 of [2];
- the kernel class (Definition 2.2): a power law in the lag with a lag-independent spatial profile;
- the framework's assignment of the Lorentz signature to the $\alpha = 2$ closure [1], which makes the causal structure at the edge time ordering only;
- the double scaling (Definition 2.5), in which the memory rate is held fixed;
- the long-wavelength scaling of the spatial kernel (Proposition 5.4), which replaces the second-order truncation used in [2, Thm. 6.1(ii)].

Given these inputs, the following are derived:

- Lemmas 4.1–4.4;
- Theorems 5.1, 5.3, 5.8 and 6.2;
- Propositions 5.4, 5.6, 6.1, 6.5, 6.7, 6.8 and 6.10, Lemma 6.11, and Corollary 5.5.

The following are not claimed:

- a value for the mass scale;
- interaction in the reciprocal sense: back-reaction, two-body coupling, entangling dynamics, or a gauge field strength generated by the carrier. These belong to the $\alpha = 2$ closure (Remark 6.6);
- Lorentz structure, which also belongs to $\alpha = 2$;
- the Lorentz action on qubit states itself, which is known [5, 6]. The new part is the reading of a carrier's internal entanglement as the time dilation of its current;
- anything about measurement [1, 4].

2. Definitions

Throughout, $\varepsilon \equiv 1 - \alpha$. Complex powers take the principal branch, and $E_c \equiv \hbar/\tau_c$.

Definition 2.1 (Causal Weyl integral and admissibility)**.** For a bounded $f : \mathbb{R} \to \mathcal{V}$ and $\alpha \in (0, 1)$, the causal (left-sided) Weyl integral is

$$W^\alpha f(t) = \frac{1}{\Gamma(\alpha)} \int_0^\infty s^{\alpha-1} f(t - s)\, ds. \tag{1}$$

It integrates over the past only. The right-sided integral $\int_t^\infty$, which some texts also call Weyl's [8], integrates over the future and is excluded.

A function f is *admissible* if its time Fourier transform, as a tempered distribution, has no support at zero frequency. On admissible functions, D^α denotes the inverse of W^α. On a carrier $e^{-iEt/\hbar}$ the symbols are $W^\alpha \mapsto (-iE/\hbar)^{-\alpha}$ and $D^\alpha \mapsto (-iE/\hbar)^\alpha$.

Definition 2.2 (Memory kernel)**.** The memory kernel is

$$k(\tau, y) = \tau_c^{-\alpha} \frac{\tau^{\alpha-1}}{\Gamma(\alpha)}\, g(y),$$

dressed by the real transport $\mathcal{P}$. Here $\tau > 0$ is the lag, and $g \geq 0$ is a normalized, isotropic spatial profile that does not depend on the lag.

Below the $\alpha = 2$ closure the causal structure is time ordering only (§6.3). So the past $J^-(x)$ at the edge is $\{t' < t\}$, and g is not required to shrink with the lag.

The *memory rate* at lag τ is $\rho_\varepsilon(\tau) \equiv (1 - \alpha)\, \tau_c^{-\alpha} \tau^{\alpha-1}/\Gamma(\alpha)$.

Definition 2.3 (Channel orientation)**.** The orientation is $\sigma \in \{+1, -1\}$. By Axiom 3 of [2], the memory channel carries the phase $e^{-i\sigma\theta}$ relative to the present channel, with $\theta = \alpha\pi/2$.

Definition 2.4 (Hilbert sector)**.** The spaces $\mathcal{V}$ and $\mathcal{H}$, the projection $\Pi_{\mathcal{H}}$ and the map f_{map} are those of [2, Defs. 2.3–2.5].

On the spinless, gauge-neutral sector $\mathcal{H} = L^2(\mathbb{R}^3)$, translation and rotation invariance make the projected memory a function of the Laplacian, $\hat{R}_{\mathcal{H}} = r_1\, \hat{g}(-i\nabla)$.

- r_1 is the coincidence weight of the fiber factors.
- $\hat{g}$ is the Fourier transform of the spatial profile, with $\hat{g}(0) = 1$ and $\hat{g}(k) = 1 - \langle r^2\rangle k^2/6 + O(k^4)$.
- Its second-order truncation is $r_0\nabla^2 + r_1$, with $r_0 = r_1\langle r^2\rangle/6$. That is the form used in [2, Thm. 6.1(ii)].

No truncation is assumed here: the kinetic term arises as a scaling limit (Proposition 5.4).

Definition 2.5 (Double scaling)**.** Take $\varepsilon \to 0^+$ with $\Lambda \equiv \varepsilon\hat{R}_{\mathcal{H}}$ held fixed.

The limiting generator $H = \sigma E_c \Lambda = \sigma\hbar\,(\varepsilon/\tau_c)\,\hat{R}_{\mathcal{H}}$ depends on ε, τ_c and $\hat{R}_{\mathcal{H}}$ only through $(\varepsilon/\tau_c)\hat{R}_{\mathcal{H}}$. So for bounded fiber factors this is the same limit as holding the memory rate ε/τ_c fixed with $\hat{R}_{\mathcal{H}}$ fixed (Proposition 6.1(2)). The two choices differ only in the approach, $O(\varepsilon)$ against $O(\varepsilon \ln \varepsilon)$. If τ_c and $\hat{R}_{\mathcal{H}}$ are both held fixed instead, the past disconnects (Proposition 6.1(1)).

The exact kinetic term needs a second scaling, of the spatial width: $\langle r^2\rangle \to 0$ with $\hbar^2/2m \equiv \varepsilon E_c r_1 \langle r^2\rangle/6$ held fixed (Proposition 5.4).

3. Axioms and the field equation

Axiom 1 (Causal accumulation law; C8 of [1])**.**

$$T(x) = \alpha J(x) - (1-\alpha)\int_{J^-(x)} k(\tau; x, x')\,\mathcal{P}(x, x')\,J(x')\,d^4x'.$$

The kernel k and the transport $\mathcal{P}$ are real. J takes values in the biquaternion carrier $\mathbb{C}\otimes\mathbb{H}$.

Remark 3.1. [2] takes J to be real. That is incompatible with its own phase channel $B\exp(iS/\hbar_{\mathrm{eff}})$, and with any solution proportional to $e^{-iEt/\hbar}$. The reality that [2, Lemma 4.3] actually uses is the reality of $\hat{R}$, and that follows from the reality of k and $\mathcal{P}$.

Axiom 2 (Two channels and channel geometry)**.** These are as in [2, Axioms 2–3]: the channels carry relative orientation $e^{i\theta}$ with $\theta = \alpha\pi/2$.

Axiom 3 (Field equation)**.** On physical configurations $T = 0$, with the memory channel oriented by Definition 2.3:

$$\alpha J(x) = (1-\alpha)\,e^{-i\sigma\alpha\pi/2}\int_{J^-(x)} k(\tau; x, x')\,\mathcal{P}(x, x')\,J(x')\,d^4x'. \tag{2}$$

Remark 3.2. Axiom 3 replaces Posit 1 of [2]. It assumes no unitarity, no generator, and no identification $\hat{H} = f_{\mathrm{map}}[\hat{T}]$.

The alternative balance law $\partial_t T = 0$ gives the same bounded admissible solutions. If J is admissible then so is T, because $\hat{T}$ is a Fourier multiplier of $\hat{J}$. A constant admissible T has no support at zero frequency, so it vanishes.

4. Unconditional lemmas

Lemma 4.1 (Weyl convergence)**.** *Let $\omega \in \mathbb{R}\setminus\{0\}$. The improper integral $\int_0^\infty s^{\alpha-1}e^{i\omega s}\,ds$ converges if and only if $0 < \alpha < 1$. When it converges, its value is $\Gamma(\alpha)(-i\omega)^{-\alpha}$. Consequently $W^\alpha e^{-iEt/\hbar} = (-iE/\hbar)^{-\alpha}e^{-iEt/\hbar}$ for $0 < \alpha < 1$. At $\alpha = 1$ the right-hand side exists only as the limit $\alpha \to 1^-$.*

Proof. Convergence for $0 < \alpha < 1$. Near $s = 0$ the integrand is integrable. On $[1, \infty)$, $s^{\alpha-1}$ decreases monotonically to 0 and $\int_1^X e^{i\omega s} ds$ is bounded, so Dirichlet's test applies.

Divergence for $\alpha \geq 1$. If $\alpha = 1$, the partial integrals $(e^{i\omega X} - 1)/(i\omega)$ have no limit. If $\alpha > 1$, the amplitude $s^{\alpha-1}$ does not tend to zero, so again there is no limit.

Value. Rotate the contour onto the ray $\arg s = \frac{\pi}{2} \operatorname{sgn} \omega$, and use Jordan's lemma for the arc. □

Lemma 4.2 (Inversion)**.** *On the sector,* $\mathcal{P}$ *is time-translation invariant and* $\hat{R}_{\mathcal{H}}$ *commutes with* W^α*. On bounded admissible* J*,* (2) *is equivalent to*

$$\alpha \, (\sigma i \tau_c)^\alpha \, D^\alpha J = \Lambda J. \tag{3}$$

Proof. On the sector, (2) reads $\alpha J = \varepsilon \, e^{-i\sigma\alpha\pi/2} \tau_c^{-\alpha} \hat{R}_{\mathcal{H}} W^\alpha J$. Apply D^α and multiply by $(\sigma i \tau_c)^\alpha = e^{i\sigma\alpha\pi/2} \tau_c^\alpha$. For the converse, apply W^α. Both maps are injective on admissible functions. □

Lemma 4.3 (Phase cancellation)**.** *Let* $g_\alpha(E) \equiv \alpha(\sigma i)^\alpha (-iE/E_c)^\alpha$ *be the symbol of the left side of* (3)*. For* $E \neq 0$*,*

$$g_\alpha(E) = \alpha |E/E_c|^\alpha \, e^{\, i\alpha\pi(\sigma - \operatorname{sgn} E)/2}.$$

For $\alpha \in (0,1) \cup (1,2)$*,* $g_\alpha(E)$ *is real if and only if* $\operatorname{sgn} E = \sigma$*, and then it is positive and strictly increasing in* $|E|$*. At* $\alpha = 1$*,* $g_1(E) = \sigma E/E_c$ *is real for both signs of* E*. Finally, setting* $f_\alpha(\lambda) \equiv \sigma E_c (\lambda/\alpha)^{1/\alpha}$ *for* $\lambda \geq 0$ *gives* $g_\alpha(f_\alpha(\lambda)) = \lambda$*.*

Proof. The principal arguments are $\arg(\sigma i) = \sigma\pi/2$ and $\arg(-iE) = -\operatorname{sgn}(E)\,\pi/2$. Hence $\arg g_\alpha = \alpha\pi(\sigma - \operatorname{sgn} E)/2 \in \{0, \pm\alpha\pi\}$, and $\pm\alpha\pi \notin \pi\mathbb{Z}$ for $\alpha \notin \{0, 1, 2\}$. The inverse relation follows by direct substitution. □

Lemma 4.4 (Operator facts)**.**

(1) $\hat{R}_{\mathcal{H}} = r_1 \hat{g}(-i\nabla)$ *is bounded and self-adjoint: it is multiplication by the real function* $r_1 \hat{g}(k)$*, and* $|\hat{g}| \leq 1$*. Its truncation* $r_0 \nabla^2 + r_1$ *is self-adjoint on* $H^2(\mathbb{R}^3)$ *and unbounded. A bounded* x*-dependent* r_1 *is a bounded perturbation, and preserves self-adjointness* [14]*.*

(2) *For the truncation used in* [2]*,* $\|\hat{R}_{\mathcal{H}}\| = \infty$*, so the spectral inclusion of* [2, Lemma 4.4] *is vacuous. For the full symbol,* $\|\hat{R}_{\mathcal{H}}\| = |r_1|$ *and the inclusion holds.*

(3) *Exchanging the kernel limit with* $\Pi_{\mathcal{H}}$ *requires strong-resolvent convergence of the projected operators* [13]*. The bound* $\|\Pi_{\mathcal{H}}\| = 1$ *alone does not suffice.*

5. Derived unitarity and the collapse

Write $\mathcal{H}_+ \equiv \operatorname{Ran} \mathbf{1}_{(0,\infty)}(\Lambda)$.

Theorem 5.1 (Bounded solutions are unitary)**.** *Fix* $\alpha \in (0,1)$ *and* σ*. Let* $J \in C_b(\mathbb{R}; \mathcal{H})$ *be admissible and solve* (3) *in the sense of* $\mathcal{H}$*-valued tempered distributions. Then:*

(i) $J(t) = e^{-it f_\alpha(\Lambda)/\hbar} \psi$ *for some* $\psi \in \mathcal{H}_+$*. Conversely, every such* J *is a solution.*

(ii) $U_\alpha(t) \equiv e^{-it f_\alpha(\Lambda)/\hbar}\big|_{\mathcal{H}_+}$ *is a strongly continuous one-parameter unitary group with self-adjoint generator* $f_\alpha(\Lambda)$*. In particular* $\|J(t)\| = \|J(0)\|$*.*

(iii) *Every frequency present satisfies* $\sigma E > 0$*, and* $\operatorname{Ran} \mathbf{1}_{(-\infty,0)}(\Lambda)$ *carries no solution.*

Proof. Step 1: fiber equation. Fourier-transform in t and pass to the spectral representation of Λ. Fiberwise, (3) becomes $\big(g_\alpha(E) - \lambda\big)\hat{J}_\lambda(E) = 0$.

Step 2: zeros. By Lemma 4.3:

- for $\lambda > 0$, the continuous function $g_\alpha - \lambda$ vanishes only at the simple zero $E_\lambda = f_\alpha(\lambda)$, where $\sigma E_\lambda > 0$;

- for $\lambda < 0$, it never vanishes;
- for $\lambda = 0$, its only zero is $E = 0$, which admissibility excludes.

Step 3: support. A tempered distribution annihilated by a continuous function is supported on that function's zero set. So $\hat{J}_\lambda$ is a finite sum of $\delta^{(n)}(E - E_\lambda)$, and $\hat{J}_\lambda = 0$ when $\lambda \le 0$. The terms with $n \ge 1$ produce growth t^n and are excluded by boundedness. Hence $J_\lambda(t) = c_\lambda e^{-iE_\lambda t/\hbar}$, which is (i).

Step 4: converse. By $g_\alpha(f_\alpha(\lambda)) = \lambda$, each such J satisfies the fiber equation. Its spectrum lies in $\sigma E > 0$, so it is admissible.

Step 5: unitarity. Functional calculus makes U_α a strongly continuous unitary group. Stone's theorem [12, 13] identifies its generator as $f_\alpha(\Lambda)$, which is self-adjoint because f_α is real and Borel. Statement (iii) is Step 2. □

Remark 5.2. Theorem 5.1 is the derived replacement for Posit 1 of [2]. The fiberwise argument of Steps 1–3 is stated at the level of rigor of the companion papers. The measurability details are standard but are not written out here.

Theorem 5.3 (Collapse to Schrödinger)**.** *Take the double scaling with Λ fixed. For every $\psi \in \mathcal{H}_+$ and every $T > 0$,*

$$\sup_{|t| \le T} \left\| U_\alpha(t)\psi - e^{-itH/\hbar}\psi \right\| \xrightarrow[\alpha \to 1^-]{} 0, \qquad H \equiv \sigma E_c \Lambda.$$

If ψ has compact spectral support in $(0, \infty)$, then

$$\left\| U_\alpha(t)\psi - e^{-itH/\hbar}\psi \right\| \le \frac{|t|}{\hbar} \left\| (f_\alpha - f_1)(\Lambda)\psi \right\|, \qquad (f_\alpha - f_1)(\lambda) = \sigma E_c\, \varepsilon\, \lambda(1 + \ln \lambda) + O(\varepsilon^2). \tag{4}$$

The limiting equation is $i\hbar\, \partial_t J = HJ$. On the spinless, gauge-neutral sector,

$$H = \sigma A\, \hat{g}(-i\nabla), \qquad A \equiv \varepsilon E_c r_1, \qquad \sigma A\, \hat{g}(k) = \sigma A - \sigma\, \frac{A\langle r^2 \rangle}{6}\, k^2 + O(k^4).$$

Positive mass with $r_1 > 0$ forces $\sigma = -1$. Then $H = -A + \frac{\hbar^2}{2m}(-\nabla^2) + O(k^4)$ with $\hbar^2/2m = A\langle r^2 \rangle/6 = \varepsilon E_c r_0$. That is the mass formula of [2, Thm. 6.1(iii)] *with its units restored. The remainder vanishes in the width scaling (Proposition 5.4).*

The stationary sector is $\mathcal{H}_+ = \{\hat{g}(k) > 0\}$. For a profile with $\hat{g} > 0$, such as a Gaussian, this is the whole sector.

Proof. Pointwise limit. Since $f_\alpha \to f_1 = \sigma E_c \lambda$ pointwise on $[0, \infty)$, the integrand $|e^{-itf_\alpha(\lambda)/\hbar} - e^{-itf_1(\lambda)/\hbar}|^2 \le 4$ tends to zero. Dominated convergence in the spectral measure of ψ gives the strong limit. Uniformity for $|t| \le T$ follows from $|e^{ia} - e^{ib}| \le \min(2, |a - b|)$ together with the local uniformity of $f_\alpha \to f_1$.

Rate. The same inequality gives the bound in (4). Expanding $\ln\big((\lambda/\alpha)^{1/\alpha}\big) = (1 + \varepsilon + \dots)(\ln \lambda + \varepsilon + \dots)$ gives the stated difference.

Identification. $f_1(\Lambda) = \sigma E_c \Lambda$, and on the sector Λ acts as multiplication by $\varepsilon r_1 \hat{g}(k)$, with $\hat{g}(k) = 1 - \langle r^2 \rangle k^2/6 + O(k^4)$ for a normalized isotropic profile. □

Proposition 5.4 (The kinetic term is a scaling limit)**.** *Let g be normalized and isotropic with finite fourth moment, and take $\sigma = -1$. Rescale the profile as $g_w(y) = w^{-3} g(y/w)$, and let $H_w = -A\, \hat{g}_w(-i\nabla)$ with $A\langle r^2 \rangle_w/6 = \hbar^2/2m$ held fixed as $w \to 0$. Then:*

1. *$H_w + A \to -\frac{\hbar^2}{2m}\nabla^2$ in the strong resolvent sense, and $e^{-it(H_w + A)/\hbar} \to e^{it\hbar\nabla^2/2m}$ strongly, uniformly for t in compact sets;*
2. *at each momentum,*

$$0 \le 1 - \frac{A\,(1 - \hat{g}_w(k))}{\hbar^2 k^2/2m} \le \frac{\langle r^4 \rangle_w}{20\, \langle r^2 \rangle_w}\, k^2 \xrightarrow[w \to 0]{} 0,$$

and for a Gaussian profile the bound is $\langle r^2\rangle_w k^2/12$*;*

(3) *the offset* $-A$*, which diverges, is a constant and is removed by a global phase;*

(4) *if* $\hat g > 0$ *everywhere, as for a Gaussian, every momentum is stationary for every* $\alpha < 1$*, so there is no energy cap. Otherwise the cap sits at the first zero of* $\hat g_w$*, which moves to infinity as* $w \to 0$*.*

Proof. For an isotropic profile, $\hat g(k) = \langle \sin(kr)/(kr)\rangle$. For real x, $x^2/6 - x^4/120 \le 1 - \sin x/x \le x^2/6$; averaging gives

$$0 \le \frac{\langle r^2\rangle k^2}{6} - \big(1 - \hat g(k)\big) \le \frac{\langle r^4\rangle k^4}{120}.$$

Multiplying by A and dividing by $\hbar^2k^2/2m = A\langle r^2\rangle k^2/6$ gives (2). Along the scaling family, $\langle r^4\rangle_w/\langle r^2\rangle_w = w^2\langle r^4\rangle/\langle r^2\rangle \to 0$.

The symbol therefore converges pointwise in k. Dominated convergence then gives the strong convergence of the resolvents and of the groups, which is (1). Part (3) is immediate. For (4), $\Lambda = \varepsilon r_1 \hat g_w(k)$ and $\hat g_w(k) = \hat g(wk)$. □

Corollary 5.5 (The skeleton and the edge are two scalings of one limit)**.** (1) *Hold* $\hat R_{\mathcal H}$ *and* τ_c *fixed, as in* [2, Lemma 4.1]*. The memory rate vanishes (Proposition 6.1(1)) and* $f_\alpha(\varepsilon \hat R_{\mathcal H}) \to 0$*. Then* $U_\alpha(t) \to \mathbf{1}$ *strongly, and* $\hbar^2/2m = \varepsilon E_c r_1 \langle r^2\rangle/6 \to 0$*, so* $m \to \infty$*. The carrier freezes. This is the kernel-blind skeleton of* [2, Thm. 5.1]*, up to its unobservable global phase.*

(2) *Hold the memory rate fixed, equivalently* Λ*. Then Theorem 5.3 applies: the memory stays live and becomes* H*.*

The "corner" and the "edge" of [2, Rem. 5.2, Table 7.4] *are therefore one limit taken in two ways.*

Proposition 5.6 (Response to a source)**.** *Let* $\alpha \in (\frac{2}{3}, 1)$*. Let* S *be a bounded source with zero past, and consider* $T = S$*, that is,* (2) *with* S *on the left. Its causal solution is* $\hat J(E) = \hat G(E)\hat S(E)$*, with*

$$\hat G(E) = \frac{(\sigma i)^\alpha (-iE/E_c)^\alpha}{g_\alpha(E) - \Lambda},$$

evaluated just above the real axis. On each spectral value $\lambda \neq 0$*,* $\hat G$ *is analytic in* $\operatorname{Im} E > 0$*; its other singularities are zeros of* $g_\alpha - \lambda$ *and the branch cut on the negative imaginary axis.*

(1) *For* $\lambda > 0$ *the zero is the real point* $E_\lambda = f_\alpha(\lambda)$*. After the source has acted, the response is the stationary solution of Theorem 5.1, with amplitude*

$$\Big|\frac{(\lambda/\alpha)\,\hat S(E_\lambda)}{g'_\alpha(E_\lambda)}\Big|,$$

plus a tail from the branch cut that decays as a power of t*.*

(2) *For* $\lambda < 0$ *the zero is at* $\operatorname{Im} E = -\pi\varepsilon|E|\big(1 + O(\varepsilon)\big) < 0$*. Such a mode exists only as a causal transient that decays at this rate. It has no bounded solution on the whole time line, consistent with Theorem 5.1(iii).*

Proof. Response function. Fourier-transform $\alpha J - \varepsilon e^{-i\sigma\alpha\pi/2}\tau_c^{-\alpha}\hat R_{\mathcal H} W^\alpha J = S$ and multiply by $(\sigma i)^\alpha(-iE/E_c)^\alpha$. This gives $\hat G$.

Analyticity. The factor $(-iE)^\alpha$, principal branch, is analytic away from $-iE \in (-\infty, 0]$, which is the negative imaginary E-axis. This analyticity in the upper half-plane is the causality of W^α.

Location of the zeros. Put $w = -iE/E_c$, so that $\operatorname{Im} E = E_c \operatorname{Re} w$, and $g_\alpha = \lambda$ becomes $w^\alpha = (\lambda/\alpha)(\sigma i)^{-\alpha}$.

- For $\lambda > 0$: $w = -i\sigma(\lambda/\alpha)^{1/\alpha}$, so $\operatorname{Re} w = 0$ and the zero is real.
- For $\lambda < 0$: the principal root has $\arg w = \pm(\pi/\alpha - \pi/2)$. A principal root exists iff $\pi - \alpha\pi/2 < \alpha\pi$, that is, $\alpha > \frac{2}{3}$. Then $\operatorname{Re} w = |w| \sin(\pi/\alpha)$, and for $\alpha = 1 - \varepsilon$ this is $-|w| \sin(\pi\varepsilon)\,(1 + O(\varepsilon))$.

Late-time response. For t after the source, close the inversion contour in the lower half-plane. The residue at the real zero gives (1). The cut gives the power-law tail [10, 9]. □

Remark 5.7. A Caputo initial-value problem with $J(0) = J_0$ corresponds to the frozen past $J \equiv J_0$ for $t < 0$. Its stationary amplitude is $\alpha^{-1}|J_0|$, not $|J_0|$. That factor belongs to the frozen past. The frozen past is not admissible for Axiom 3, so the factor is not a prediction of the field equation.

Theorem 5.8 (The limit is one-sided)**.** (1) Convergence. *W^α converges on oscillating carriers only for $\alpha < 1$ (Lemma 4.1). The memory term of (2) at $\alpha = 1$ exists only as the limit from below.*

(2) Sign. *With σ fixed, the memory-generated coefficient $\hbar^2/2m = -\sigma\varepsilon E_c r_1 \langle r^2 \rangle/6$ is odd in the memory rate $\varepsilon E_c/\hbar$. Positive mass can hold on one side of $\alpha = 1$ only.*

(3) No continuation above $\alpha = 1$. *For $\alpha \in (1, 2)$, take the source problem with zero past. There the Riemann–Liouville integral is finite, and $\hat{G}$ is defined by the continued symbol. With $r_1 > 0$ and $\varepsilon < 0$, every mode with $\hat{g}(k) > 0$ has $\lambda < 0$; for a Gaussian profile that is every mode. For such a mode, $g_\alpha - \lambda$ has a zero with $\operatorname{Im} E \approx +\pi|\varepsilon||E| > 0$, in the upper half-plane. Then:*

- *the causal response, obtained by inverting above every singularity, grows at that rate;*
- *the bounded response, obtained by inverting on the real axis, is nonzero before the source acts.*

So the one-way edge equation has no causal, bounded continuation past $\alpha = 1$.

Proof. Part (1) is Lemma 4.1. Part (2) is immediate from Theorem 5.3. For part (3), the argument in the proof of Proposition 5.6 gives $\operatorname{Re} w = |w| \sin(\pi/\alpha)$, which is positive for $\alpha \in (1, 2)$. A zero of the response function in the upper half-plane rules out a response that is both causal and bounded. □

Remark 5.9 (What part (3) does and does not say). Part (3) concerns the equation of Axiom 3 continued above $\alpha = 1$ with its orientation fixed. It is not a statement about the framework's dynamics on $(1, 2]$, which are set by the reciprocal $\alpha = 2$ closure. In the convention of [3, Eq. (60)], $\alpha \geq 1$ is where the history enters with nonnegative weight.

The upper-half-plane zero has the same mathematical form as in the Abraham–Lorentz–Dirac equation, but no third derivative is involved. One reading consistent with the framework: one-way transport is consistent only up to $\alpha = 1$, and beyond it the return leg must be present.

6. The two Markov limits, causal structure, interaction, and readout

The framework has two Markov limits [3, Sec. 1].

- In the *forgetful* (Dirac) limit the kernel concentrates at zero lag, and nothing of the past enters.
- In the *faithful* (light-cone) limit the kernel retains every contribution of the past distinctly, so the present state carries the whole history. A closed system, with conserved mass and energy, cannot reach it [3, Sec. 2].

This section shows three things.

- A Dirac memory kernel on the carrier cannot give unitary evolution.

- The edge is Markov through the fundamental theorem of calculus instead. The carrier is the flat accumulation of the past, and the law for its rate is local. That keeps the accumulated sum, not each contribution, so it is not the faithful limit, and it needs neither records nor a larger state space.
- The corner $\alpha = 1$ at fixed time unit is different again: there the past simply disconnects.

The section then records the structure of the edge: it is Galilean, it carries no interaction, and its readout survives tensor composition.

6.1. **The corner and the accumulator.**

Proposition 6.1 (Memory rate)**.** $\rho_\varepsilon(\tau) = \varepsilon\, \tau_c^{-1+\varepsilon} \tau^{-\varepsilon}/\Gamma(1-\varepsilon)$.

(1) *With τ_c fixed, $\rho_\varepsilon \to 0$ uniformly on compact lag intervals. The past disconnects even from below, and the edge degenerates to the skeleton (Corollary 5.5(1)).*
(2) *With $\tau_c = \varepsilon/\kappa$, $\rho_\varepsilon(\tau) \to \kappa$ for every fixed $\tau > 0$. The kernel becomes the flat accumulator of rate κ, the accumulator of the fundamental theorem of calculus. It weights every contribution equally but keeps only their sum, so it is neither a δ nor the faithful limit.*

Proof. For (1), $\rho_\varepsilon \le C\varepsilon\, \tau^{-\varepsilon}$ on compacts. For (2), $\rho_\varepsilon = \kappa^{1-\varepsilon}\varepsilon^\varepsilon \tau^{-\varepsilon}/\Gamma(1-\varepsilon)$, and $\varepsilon^\varepsilon \to 1$, $\tau^{-\varepsilon} \to 1$, $\Gamma(1-\varepsilon) \to 1$. □

6.2. **A Dirac memory kernel: no stationary band.**

Theorem 6.2 (Normalizable kernels have no stationary band)**.** *Replace the power law in (2) by a lag weight $w \ge 0$ with $\int_0^\infty w = 1$, of any width. This includes every family $w \to \delta$.*

On a spectral value λ of $\hat{R}_{\mathcal{H}}$, a plane wave $e^{-iEt/\hbar}$ with real E solves the field equation only if $\varepsilon|\lambda| \ge \alpha$. Hence on $\{\varepsilon|\lambda| < \alpha\}$ the field equation has no bounded solution on the whole time line. That subspace contains every small value of $\Lambda = \varepsilon\lambda$, and so the entire low-energy band.

Proof. On a plane wave the field equation reads $\alpha = \varepsilon\, e^{-i\sigma\theta}\lambda\, \hat{w}(E)$, where $\hat{w}(E) = \int w(\tau) e^{iE\tau/\hbar} d\tau$. For real E, $|\hat{w}(E)| \le \int w = 1$, so a real root needs $\alpha \le \varepsilon|\lambda|$. The support argument of Theorem 5.1 (Steps 1–3) then excludes bounded solutions wherever no real root exists. □

Example 6.3 (A narrow exponential kernel)**.** Take $w(\tau) = e^{-\tau/\tau_w}/\tau_w$, so that $\hat{w}(E) = 1/(1 - iE\tau_w/\hbar)$. The root is exact:

$$E = -\frac{i\hbar}{\tau_w}\Big[1 - \frac{\varepsilon\lambda}{\alpha} e^{-i\sigma\theta}\Big].$$

Hold the would-be generator $\kappa \equiv \varepsilon\lambda\hbar/\tau_w$ fixed. Then

$$\operatorname{Re} E = \sigma\frac{\kappa}{\alpha}\sin\theta \to \sigma\kappa, \qquad \operatorname{Im} E = -\frac{\hbar}{\tau_w} + \frac{\kappa}{\alpha}\cos\theta.$$

The real part is the Schrödinger frequency. But every mode decays at the inverse kernel width, which becomes infinite as $\tau_w \to 0$ (Table 7). A narrow kernel's symbol is nearly real, so it cannot absorb the channel factor $e^{-i\sigma\theta} \to \mp i$, which therefore appears as damping rather than as a phase.

Remark 6.4 (Why the power law). The power-law kernel is not normalizable, since $\int_0^\infty \tau^{\alpha-1} d\tau = \infty$. Its symbol $(-iE\tau_c/\hbar)^{-\alpha}$ is unbounded as $E \to 0$. So |symbol| $= \alpha/(\varepsilon\lambda)$ is met at a finite real energy $E = E_c(\varepsilon\lambda/\alpha)^{1/\alpha}$, which is the stationary band of Theorem 5.1. The symbol's phase is exactly the channel angle $\alpha\pi/2$, which is how [2, Def. 2.2] defines the channel angle.

Infinite total memory weight, multiplied by a vanishing ε, gives the finite rate of Proposition 6.1(2). The edge is therefore Markov through the fundamental theorem of calculus, reached

from a law with memory (power law, full width). A Dirac memory kernel damps; the accumulator is unitary (Theorem 5.1). In the table of [2], the Schrödinger form belongs in the full-width (power-law) column at 1^-, not in the δ column. Posit 1 was covering the δ column's lack of a stationary band.

Proposition 6.5 (Markov without enlarging the state space)**.** *For $0 < \alpha < 1$ the power-law kernel is a superposition of exponentials,*

$$\frac{\tau^{\alpha-1}}{\Gamma(\alpha)} = \int_0^\infty e^{-s\tau}\mu_\alpha(s)\,ds, \qquad \mu_\alpha(s) = \frac{\sin\pi\alpha}{\pi}\,s^{-\alpha},$$

so $W^\alpha J = \int_0^\infty \mu_\alpha(s)\,y_s\,ds$, with auxiliary modes obeying $\dot y_s = -s\,y_s + J$. A Markov description of the memory at $\alpha < 1$ therefore needs a continuum of auxiliary modes, which is an enlarged state space.

As $\alpha \to 1^-$ the weight μ_α concentrates at $s = 0$: on any interval $[0, S]$, the fraction of its mass in $[0, \delta]$ is $(\delta/S)^{1-\alpha} \to 1$. The continuum collapses onto the single mode $y_0 = \int^t J$, which is the accumulated state itself, and the enlargement disappears.

Proof. Use $\int_0^\infty s^{-\alpha}e^{-s\tau}ds = \Gamma(1-\alpha)\tau^{\alpha-1}$ and $\Gamma(\alpha)\Gamma(1-\alpha) = \pi/\sin\pi\alpha$. Each $y_s(t) = \int_{-\infty}^t e^{-s(t-t')}J(t')\,dt'$ satisfies $\dot y_s = -sy_s + J$. The mass fraction is $\int_0^\delta s^{-\alpha}ds \big/ \int_0^S s^{-\alpha}ds = (\delta/S)^{1-\alpha}$. □

So the edge needs neither records nor extra state. The faithful limit, which would keep every contribution distinct, is not what it uses.

6.3. **Causal structure: the edge is Galilean.** In the framework the Lorentz signature appears with the quadratic closure $JJ^\dagger$ at $\alpha = 2$, through the invariant $\det(JJ^\dagger) = |\det J|^2$ [1]. The same closure gives the Born readout through the trace, $\frac{1}{2}\mathrm{tr}(JJ^\dagger) = \sum_a |Z_a|^2$. That trace is a density, the time component of a current; it is not a Lorentz scalar. Below the closure there is no determinant, so there is no metric signature and no light cone.

At the $\alpha \to 1^-$ edge the causal structure is therefore time ordering alone: $J^-(x)$ reduces to the time-ordered past $\{t' < t\}$, with no bound on spatial reach. This is the Galilean structure of the Schrödinger equation, whose free propagator is nonzero at every distance for every $t > 0$. Three things follow.

(1) The lag-independent spatial profile of Definition 2.2 is admissible, and so is shrinking its width (Proposition 5.4).
(2) Instantaneous spatial reach in a δ route is admissible as well. That route fails on unitarity (Theorem 6.2), not on causality.
(3) A light cone imposed at the edge imports $\alpha = 2$ structure. For reference only, Appendix C records what imposing one would give.

6.4. **The edge carries no interaction.**

Remark 6.6 (Single legs). In (2) the past enters only through transports $\mathcal{P}(x, x')$ from a point of the past to the present, each acting within one carrier. Hence:

(1) *No coupling between carriers.* For two carriers the edge operator is a sum of single-carrier terms, so the evolution is $U_A \otimes U_B$. That preserves the Schmidt coefficients between the carriers: it can carry entanglement that already exists, but it cannot create any. The reason is the transport structure, not linearity. A linear equation with a two-carrier coupling would entangle.

(2) *A coupling would need the return leg.* A term transferring from A to B alone is not Hermitian. Keeping $\hat{R}_{\mathcal{H}}$ self-adjoint, and with it the unitarity of Theorem 5.1, requires its adjoint: the return leg from B to A. For a unitary transport $P_\gamma^\dagger = P_{\gamma^{-1}}$, so that adjoint is the dagger of the closure $JJ^\dagger$. Reciprocal structure of this kind belongs to the $\alpha = 2$ closure.
(3) *No holonomy from the carrier.* Holonomy needs a closed path [1]. What the edge can carry is a fixed background acting one way, through a bounded $r_1(x)$ (Lemma 4.4) or a prescribed transport. That is a test particle that does not act back.

This agrees with the restriction of [2, Thm. 6.1] to the free form: that restriction is structural, not a matter of convenience.

6.5. **Readout: one carrier, and composites.**

Proposition 6.7 (One carrier is two factors)**.** *Read $J \in M_2(\mathbb{C})$ as a vector $|J\rangle\!\rangle = \sum_{ij} J_{ij}|i\rangle|j\rangle \in \mathbb{C}^2 \otimes \mathbb{C}^2$. The row index is the left factor, on which Lorentz transformations act ($J \to LJ$). The column index is the right factor, on which the internal phase acts ($J \to JU$). Then:*

(1) *$JJ^\dagger = \mathrm{Tr}_{\mathrm{right}}|J\rangle\!\rangle\langle\!\langle J|$, the reduced state of the left factor;*
(2) *J is a product state $\iff$ $\operatorname{rank} J = 1$ $\iff$ $\det J = 0$, which is a null closure* [3, Eq. (40)]*;*
(3) *the eigenvalues of $JJ^\dagger$ are the Schmidt weights: their sum $\operatorname{tr} JJ^\dagger$ is the Born normalization, and their product $\det JJ^\dagger = |\det J|^2$ is the Lorentzian norm;*
(4) *the concurrence of the normalized state is*

$$C = \frac{2|\det J|}{\operatorname{tr}(JJ^\dagger)} = \sqrt{1 - v^2/c^2} = \frac{1}{\gamma},$$

where $v/c = |\vec{x}|/x^0$ is the speed of the closure $JJ^\dagger = x^0 I_2 + \vec{x}\cdot\vec{\sigma}$.

Proof. For (1): $\left(\mathrm{Tr}_{\mathrm{right}}|J\rangle\!\rangle\langle\!\langle J|\right)_{ik} = \sum_j J_{ij}\bar{J}_{kj}$. For (2): a 2×2 matrix has rank one iff its determinant vanishes, and a rank-one matrix is a product $\psi\eta^\dagger$. Part (3) is the singular-value decomposition of J. For (4): the eigenvalues of $x^0 I_2 + \vec{x}\cdot\vec{\sigma}$ are $x^0 \pm |\vec{x}|$, and for a pure bipartite state $C = 2\sqrt{p_+ p_-}$ with $p_\pm = (x^0 \pm |\vec{x}|)/(2x^0)$. □

So "Born by trace" and "Lorentz by determinant" are the sum and the product of the Schmidt weights of a single carrier. A carrier at rest is maximally entangled between its factors; a massless carrier is a product. For a massive carrier the left closure $JJ^\dagger$ is therefore mixed. The pure state read on the left is $(J\eta)(J\eta)^\dagger$ for a reference leg η, as in [3, Eq. (45)].

Proposition 6.8 (Lorentz action on the carrier's entanglement)**.** *Let $G \in SL(2,\mathbb{C})$ act on the left factor, $J \to GJ$. Then $Q \to GQG^\dagger$. This is the proper orthochronous Lorentz action on the Hermitian slice* [3, Eqs. (35)–(36)]*, and:*

(1) *$|\det J|$, the invariant density, is unchanged, while $x^0 = \frac{1}{2}\operatorname{tr} Q$, the density in the frame, is not;*
(2) *rotations, $G \in SU(2)$, are local unitaries and leave the concurrence unchanged; boosts, with G Hermitian and positive, are invertible local filtering operations and change it* [5]*;*
(3) *writing the speed of the closure as $\beta = \tanh\eta$, with η its rapidity,*

$$C = \operatorname{sech}\eta, \qquad p_\pm = \frac{e^{\pm\eta}}{2\cosh\eta}, \qquad \frac{p_+}{p_-} = e^{2\eta},$$

so the Schmidt weights are proportional to the relativistic Doppler factors $e^{\pm\eta}$. Collinear boosts add rapidities: boosting a carrier at rest by η_1 and then by η_2 along one axis gives $C = \operatorname{sech}(\eta_1 + \eta_2)$;

(4) *the pure states of the left factor, which are the null closures, form* $\mathbb{CP}^1$*, the celestial sphere. On it* $SL(2,\mathbb{C})$ *acts by Möbius transformations* [7]*; see also* [3, App. A.2].

Proof. $\det(GJ) = \det J$ when $\det G = 1$, and the congruence $Q \to GQG^\dagger$ preserves $\det Q = x_\mu x^\mu$. For $G \in SU(2)$ the trace is also preserved, so $C = 2|\det J|/\operatorname{tr} Q$ is unchanged.

For (3), use Proposition 6.7(4) with $\beta = \tanh\eta$: then $\sqrt{1-\beta^2} = \operatorname{sech}\eta$ and $p_\pm = (1 \pm \beta)/2$. A boost of rapidity η along a unit vector n is $e^{\eta\, n\cdot\sigma/2}$. Acting on a carrier at rest, where $Q = \rho I$, it gives $Q = \rho\, e^{\eta\, n\cdot\sigma}$, with $x^0 = \rho\cosh\eta$ and $|\vec{x}| = \rho\sinh\eta$. Collinear exponents add.

Part (4) is the standard spinor description of null directions. □

Remark 6.9 (What is known, and what is read). The Lorentz structure of the qubit state space is known. The Bloch ball is the trace-one section of the future light cone of positive 2×2 matrices [6], and local filtering on a qubit acts as a Lorentz transformation [5].

What the carrier adds is a reading: the entanglement between a carrier's two factors is the time dilation of its current. A carrier at rest is maximally entangled, a massless carrier is a product state, and boosts move between the two as $\operatorname{sech}\eta$.

This is a statement about the closure $JJ^\dagger$, which is an $\alpha = 2$ object. It does not conflict with §6.3: the dynamics of the edge is Galilean, and the Lorentz group here acts on the carrier's closure, where [3] places the Lorentzian structure, not on the spacetime of the edge.

Proposition 6.10 (Readout under composition)**.** *Let* $J \in M_N(\mathbb{C})$ *have any rank, let* $Q = JJ^\dagger$*, and let* P_m *be projectors with* $\sum_m P_m = I_N$*. Then* $p_m = \operatorname{Tr}(P_m Q)/\operatorname{Tr} Q$ *is a probability distribution. For composite carriers* $J_{AB} \in M_2(\mathbb{C}) \otimes M_2(\mathbb{C}) \cong M_4(\mathbb{C})$*, and likewise* $M_{16}(\mathbb{C})$ *for four carriers:*

(1) *product carriers give product statistics: if* $J_{AB} = J_A \otimes J_B$ *then* $Q_{AB} = J_A J_A^\dagger \otimes J_B J_B^\dagger$ *and* $p(m,n) = p_A(m)\, p_B(n)$*;*
(2) *marginals are consistent for every* J_{AB}*:* $\operatorname{Tr}\big((P_m \otimes I)Q_{AB}\big) = \operatorname{Tr}\big(P_m \operatorname{Tr}_B Q_{AB}\big)$*;*
(3) *the rank of* J_{AB} *and the entanglement between* A *and* B *are different things. The rank measures entanglement between the left and right halves of* J_{AB}*, and decides whether* Q_{AB} *is pure. Entanglement between* A *and* B *is a property of* Q_{AB}*. For example,* $J_{AB} = \Phi^+\eta^\dagger$ *has rank one and gives a pure* Q_{AB} *with* $\operatorname{Tr}_B Q_{AB} = I_2/2$*: maximally entangled between* A *and* B*.*

Proof. Positivity and $\sum_m P_m = I$ give a distribution. For (1), $(J_A \otimes J_B)(J_A \otimes J_B)^\dagger = J_A J_A^\dagger \otimes J_B J_B^\dagger$, and the trace of a tensor product is the product of the traces. Part (2) is the definition of the partial trace, and (3) is direct computation. □

Tensor composition therefore changes the available states and interactions, not the probability rule.

Lemma 6.11 (Intrinsic normalization above M_2)**.** *Dividing by* $\operatorname{Tr} Q$ *needs no rank condition. The stronger claim is that the closure normalizes itself:* $2J = JJ^\dagger$ *gives a projector* $P = J/2$ *with* $\operatorname{tr} P = 1$ [1]*. That needs* J *of rank one.*

- *In* $M_2(\mathbb{C})$*, rank one is the condition* $\det J = 0$*.*
- *In* $M_N(\mathbb{C})$ *with* $N \geq 3$*, rank one is* $\Lambda^2 J = 0$*, the vanishing of all* 2×2 *minors. This is still quadratic in* J*. The condition* $\det J = 0$ *alone is not enough.*

Proof. $2J = JJ^\dagger$ forces $J = J^\dagger$ and $J^2 = 2J$, so $P = J/2$ is a projector with $\operatorname{tr} P = \operatorname{rank} J$. Rank at most one is equivalent to $\Lambda^2 J = 0$. For $N \geq 3$, $J = \operatorname{diag}(2, 2, 0, \ldots, 0)$ satisfies $2J = JJ^\dagger$ and $\det J = 0$, but has $\operatorname{tr} P = 2$. □

7. Numerical verification

All statements above were run with the script `scripts/cal_schrodinger_collapse.py` in the repository https://github.com/JewJitsu11B/Liorhybrid. It uses NumPy only and runs in about ten seconds. Units are $\hbar = \tau_c = 1$, so $E_c = 1$. Methods are described in Appendix B.

Bounded solutions, with $\sigma = -1$ and the full Gaussian symbol $\hat{g}(k) = e^{-\langle r^2\rangle k^2/6}$. Take $\langle r^2\rangle = 0.75$ and $A = 4$, so that $\hbar^2/2m = 1/2$. Every mode is stationary. The test state is a Gaussian packet with $k_0 = 1$ and width 2.

ε	$\max\lvert E_\alpha - E_1\rvert$	$\varepsilon \max \lambda\lvert 1 + \ln\lambda\rvert$	ratio	phase-aligned error, $t \le 40$	norm drift
10^{-1}	1.246	0.954	1.305	1.41	6×10^{-16}
10^{-2}	9.78×10^{-2}	9.55×10^{-2}	1.025	2.87×10^{-1}	4×10^{-16}
10^{-3}	9.57×10^{-3}	9.55×10^{-3}	1.002	2.83×10^{-2}	6×10^{-16}
3×10^{-4}	2.87×10^{-3}	2.86×10^{-3}	1.001	8.47×10^{-3}	6×10^{-16}

Table 1. Theorems 5.1–5.3. The energy error matches the first-order term of (4), and the norm is conserved to machine precision. The raw packet error is dominated by a global phase drift from the constant term. With that phase removed, the error is linear in ε, at about $28\,\varepsilon$ over $t \le 40$.

$\langle r^2\rangle$	offset $-A$	$k = 0.5$	$k = 1$	$k = 2$	$k = 4$	bound at $k = 4$
1	-3	2.05×10^{-2}	7.89×10^{-2}	0.270	0.651	1.33
0.1	-30	2.08×10^{-3}	8.29×10^{-3}	3.26×10^{-2}	0.122	0.133
0.01	-300	2.08×10^{-4}	8.33×10^{-4}	3.33×10^{-3}	1.32×10^{-2}	1.33×10^{-2}
0.001	-3000	2.08×10^{-5}	8.33×10^{-5}	3.33×10^{-4}	1.33×10^{-3}	1.33×10^{-3}

Table 2. Proposition 5.4: relative error of the kinetic term, $\lvert A(1 - \hat{g}(k))/(\hbar^2k^2/2m) - 1\rvert$, for a Gaussian profile with $A\langle r^2\rangle/6 = \hbar^2/2m = 1/2$ held fixed. It stays within the bound $\langle r^2\rangle k^2/12$ and vanishes as the width shrinks. The offset $-A$ diverges, but only as a global phase.

Source-driven runs. The integral equation $T = S$ is time-stepped directly, with Weyl integration over a zero past and the full history kept. The source is a Gaussian pulse at $t_0 = 10$ of width 2, tuned to the mode. No initial-value data enter.

α	h	mean $\lvert J\rvert$, $t \in [80, 120]$	residue prediction	ratio
0.900	0.02	6.95236	6.95787	0.99921
0.900	0.01	6.95674	6.95787	0.99984
0.990	0.01	5.16659	5.16724	0.99987
0.999	0.01	5.02771	5.02833	0.99988
1.000	0.01	5.01264	5.01326	0.99988
1.010	0.01	4.86571	4.86630	0.99988

Table 3. Modes with $\Lambda = +1$, Proposition 5.6(1). After the pulse the mode oscillates forever at the amplitude of the residue at the real zero. This is the stationary solution of Theorem 5.1, excited causally. The ratio converges as h is halved.

The two Markovian regimes.
Readout.

α	zero of $g_\alpha - \Lambda$	$-\operatorname{Im} E$	$\pi\varepsilon\lvert E\rvert$	measured decay rate
0.970	$1.027032 - 0.100104\,i$	$+1.0010 \times 10^{-1}$	$+9.73 \times 10^{-2}$	$+1.0007 \times 10^{-1}$
0.990	$1.009695 - 0.032052\,i$	$+3.2052 \times 10^{-2}$	$+3.17 \times 10^{-2}$	$+3.2049 \times 10^{-2}$
0.999	$1.000997 - 0.003148\,i$	$+3.1479 \times 10^{-3}$	$+3.14 \times 10^{-3}$	$+3.1478 \times 10^{-3}$
1.000	1.000000	0	0	$+2.5 \times 10^{-7}$
1.001	$0.998997 + 0.003135\,i$	-3.1353×10^{-3}	-3.14×10^{-3}	-3.1348×10^{-3}
1.010	$0.989718 + 0.030795\,i$	-3.0795×10^{-2}	-3.11×10^{-2}	-3.0792×10^{-2}
1.030	$0.967645 + 0.088790\,i$	-8.8790×10^{-2}	-9.16×10^{-2}	-8.8782×10^{-2}

TABLE 4. Modes with $\Lambda = -1$, Proposition 5.6(2) and Theorem 5.8(3). Below $\alpha = 1$ such modes exist only where $\hat{g}$ changes sign; above $\alpha = 1$, where $\varepsilon < 0$, they are the whole sector. Below $\alpha = 1$ the zero lies in $\operatorname{Im} E < 0$, giving causal decay. At $\alpha = 1$ it lies on the real axis. Above, it lies in $\operatorname{Im} E > 0$, and the causal response grows. The measured rate is $(\ln|J(40)| - \ln|J(80)|)/40$.

α	$\lvert J(t_0 - 10)\rvert$	$\lvert J(t_0 - 20)\rvert$	$\lvert J(t_0 + 40)\rvert$
0.99	4.0×10^{-6}	7.5×10^{-11}	1.44
1.01	3.58	2.63	4.2×10^{-6}

TABLE 5. Bounded responses, from inversion on the real axis, for $\Lambda = -1$. From below the bounded response is causal: what appears before the pulse is only the Gaussian's own tail. Above $\alpha = 1$, keeping the response bounded makes it precede its source (Theorem 5.8(3)).

	$\tau_c = 1$ fixed		$\tau_c = \varepsilon$		
ε	$\tau = 1$	$\tau = 100$	$\tau = 0.01$	$\tau = 1$	$\tau = 100$
10^{-1}	9.36×10^{-2}	5.90×10^{-2}	1.178	0.743	0.469
10^{-2}	9.94×10^{-3}	9.49×10^{-3}	0.994	0.949	0.907
10^{-3}	9.99×10^{-4}	9.95×10^{-4}	0.997	0.993	0.988
10^{-4}	1.00×10^{-4}	1.00×10^{-4}	1.000	0.999	0.999

TABLE 6. Memory rate $\rho_\varepsilon(\tau)$, Proposition 6.1. With τ_c fixed the rate is about ε, and the past disconnects even from below. With $\tau_c \propto \varepsilon$ the rate tends to 1 at every lag: a flat accumulator, not a δ.

ε	τ_w	$\operatorname{Re} E$	$\operatorname{Im} E$	$-1/\tau_w$
10^{-2}	1	$+1.0100$	-0.984	-1
10^{-2}	0.01	$+1.0100$	-99.98	-100
10^{-3}	1	$+1.0010$	-0.998	-1
10^{-3}	0.1	$+1.0010$	-9.998	-10
10^{-3}	0.001	$+1.0010$	-1000.0	-1000

TABLE 7. The normalizable exponential kernel of Example 6.3, with generator $\kappa = 1$ and $\sigma = +1$. The real part reproduces the Schrödinger frequency, but every mode decays at about $1/\tau_w$. The power law with the same κ gives the real root $1.0010 + 0i$ at $\varepsilon = 10^{-3}$.

Check	Result
$\|\mathrm{Tr}_{\mathrm{right}}\|J\rangle\rangle\langle\langle J\| - JJ^\dagger\|$, 200 random J	1.8×10^{-15}
$\|C - 1/\gamma\|$, same J	1.2×10^{-15}
trace readout against $\|\Phi_m^\dagger \Psi\|^2/\Psi^\dagger\Psi$, $N \in \{2, 4, 16, 1024\}$	3.7×10^{-16}
product carriers: $\|p(m,n) - p_A(m)p_B(n)\|$	1.1×10^{-16}
rank-4 J_{AB}: marginal mismatch	0
$J_{AB} = \Phi^+\eta^\dagger$: rank, $\mathrm{Tr}_B Q$	1, $I_2/2$
$\mathrm{diag}(2, 2, 0, 0)$: $\det J$, largest 2×2 minor, $\mathrm{tr}\, P$	0, 4, 2

TABLE 8. Propositions 6.7–6.10 and Lemma 6.11.

η	C	$\mathrm{sech}\,\eta$	p_+/p_-	$e^{2\eta}$	$\|\det J\|$
0	1.000000	1.000000	1.000000	1.000000	2.0000
0.5	0.886819	0.886819	2.718282	2.718282	2.0000
1	0.648054	0.648054	7.389056	7.389056	2.0000
2	0.265802	0.265802	54.59815	54.59815	2.0000

TABLE 9. Proposition 6.8: a carrier at rest, $J = \sqrt{2}\,U$ with $U \in SU(2)$, boosted by rapidity η along a random axis. A rotation changes C by 1.1×10^{-16}. Boosts of 0.7 and then 0.9 along one axis give $C = 0.387978 = \mathrm{sech}\, 1.6$.

8. Remarks on scope

Remark 8.1 (Status ledger). *Inputs:*

- the field equation (Axiom 3);
- the power-law kernel class (Definition 2.2);
- time ordering as the causal structure below the $\alpha = 2$ closure;
- the double scaling at fixed memory rate (Definition 2.5);
- the long-wavelength scaling of the spatial kernel (Proposition 5.4).

Derived from these: every lemma, theorem, proposition and corollary of §§4–6.
Not claimed:

- a value for the mass scale;
- interaction, entanglement, Lorentz structure, or a gauge field strength generated by the carrier, all of which belong to $\alpha = 2$;
- any initial-value (Caputo) statement, since a frozen past is not admissible.

Remark 8.2 (What changes relative to [2]).

- Posit 1, and Theorem 6.1(i) built on it, are replaced by Axiom 3 together with Theorems 5.1 and 5.3.
- The δ column at the 1^- edge has no stationary band once the channel phase is used (Theorem 6.2). That is the gap Posit 1 was covering. The Schrödinger form belongs in the power-law (full-width) column.
- Below $\alpha = 2$ the causal structure is Galilean, so the δ route's instantaneous spatial reach is admissible. Its obstruction is unitarity, not causality.
- The free form of Theorem 6.1(ii) is structural: the edge carries no interaction (Remark 6.6).
- The second-order form of Theorem 6.1(ii) is the long-wavelength truncation of $r_1\hat{g}(-i\nabla)$. Keeping the full symbol and shrinking the width gives the Schrödinger kinetic term exactly, with no energy cap (Proposition 5.4).
- Axiom 1 should require k and $\mathcal{P}$ to be real, not J.
- In Lemma 4.3, real plus symmetric gives a symmetric operator; self-adjointness is Lemma 4.4(1).
- In Lemma 4.4, $\|\hat{R}\| = \infty$, and exchanging the limit with the projection needs strong-resolvent convergence.
- The mass formula of Theorem 6.1(iii) holds at fixed memory rate. With τ_c fixed the mass diverges (Corollary 5.5(1)).
- The constant shift in Eq. (11) is inert only at $\alpha = 1$. For $\alpha < 1$ it sets the stationary sector, by Theorem 5.1(iii).
- The 1^- edge of Table 7.4 is one-sided by Theorem 5.8, not by convention.

Remark 8.3 (Dynamics is memory). Corollary 5.5 and Theorem 6.2 make [2, Rem. 7.3] precise.

- If the memory weight vanishes at fixed time unit, the past disconnects and the carrier freezes.
- If it vanishes at fixed memory rate, the memory becomes the Hamiltonian. That needs a memory of infinite total weight, which accumulates the past as a sum rather than keeping each contribution. A normalizable memory, however narrow, cannot supply a unitary generator.

The Schrödinger generator is the image of an infinite accumulated past under the double scaling, transported one way, with no return leg.

Appendix A. Notation

α, $\varepsilon = 1-\alpha$	fractional order and memory weight
σ, $\theta = \alpha\pi/2$	channel orientation and channel angle
W^α, D^α	causal Weyl integral and its inverse (Definition 2.1)
τ_c, $E_c = \hbar/\tau_c$	memory time unit and its energy
$\rho_\varepsilon(\tau)$	memory rate at lag τ (Definition 2.2)
g, $\hat{g}$, $\langle r^2\rangle$	spatial profile, its Fourier transform, its second moment
$\hat{R}_\mathcal{H} = r_1\hat{g}(-i\nabla)$	projected memory operator ($r_0\nabla^2 + r_1$ is its truncation)
$A = \varepsilon E_c r_1$	energy scale of the memory; $\hbar^2/2m = A\langle r^2\rangle/6$
$\Lambda = \varepsilon\hat{R}_\mathcal{H}$	double-scaled memory operator
g_α, f_α	symbol of the left side of (3), and stationary energy
$\mathcal{H}_+$	$\operatorname{Ran}\mathbf{1}_{(0,\infty)}(\Lambda)$, the stationary sector
w, $\hat{w}$, τ_w	normalizable lag weight, its symbol, its width
$\lvert J\rangle\rangle$	the carrier read as a vector in $\mathbb{C}^2\otimes\mathbb{C}^2$
$Q = JJ^\dagger$, P_m	positive closure; outcome projectors
C, γ	concurrence; Lorentz factor of the closure
$\beta = \tanh\eta$	speed and rapidity of the closure

Appendix B. Numerical methods

Time stepping. The Weyl integral over a zero past is discretized by BDF2 convolution quadrature [11], with weights q_j generated by $\big((1-\zeta)+(1-\zeta)^2/2\big)^{-\alpha}$. The full history is kept. Each step solves

$$(\alpha - \kappa h^\alpha q_0)\, J_n = S_n + \kappa h^\alpha \sum_{j\ge 1} q_j J_{n-j}, \qquad \kappa = \Lambda e^{-i\sigma\alpha\pi/2}.$$

This is the integral field equation itself. No derivative and no initial-value data are used.

Zeros and residues. Zeros of $g_\alpha - \Lambda$ are found by Newton iteration. Residue amplitudes use $\hat{S}(E) = \sqrt{2\pi}\, w\, e^{-w^2(E-\Omega)^2/2} e^{i(E-\Omega)t_0}$. The bounded responses come from direct inversion on the real axis, on a grid of 1.6×10^5 points over $|E|\le 40$.

Packets. Packets use a periodic grid of 2048 points on a box of length 400, with a unitary FFT. Modes with amplitude below 10^{-10} of the maximum are dropped.

Markov limits. The memory rate is evaluated in closed form. The exponential-kernel roots are the closed form of Example 6.3.

Readout. Partial traces are taken by index contraction. Outcome bases are random unitaries from QR factorization, and the minors are enumerated directly. Boosts are $e^{\eta\, n\cdot\sigma/2}$ along a random axis, applied to a carrier at rest $\sqrt{2}\,U$ with $U\in SU(2)$ random.

Appendix C. For reference: a light cone imposed by hand

The framework assigns the Lorentz signature to the $\alpha = 2$ closure (§6.3). Nothing in this appendix describes the $\alpha\to 1^-$ edge. It records what follows if a light cone is nevertheless imposed on the edge, as one ingredient of an eventual crossover to $\alpha = 2$.

Setting. Let the spatial profile depend on the lag and be supported in $|y|\le c\tau$. Take it self-similar, so that it is a superposition of thin shells of speed $u\in[0,c]$ with weight $w(u)$, and set $v_{\text{eff}}^2 = \int u^2 w(u)\, du$. Hold $E_0 \equiv \varepsilon E_c r_1$ fixed. On a plane wave the field equation reads

$$\alpha = \varepsilon r_1 e^{-i\sigma\alpha\pi/2}\tau_c^{-\alpha} m_\alpha(E,k), \qquad m_\alpha = \frac{1}{\Gamma(\alpha)}\int_0^\infty \tau^{\alpha-1} e^{iE\tau/\hbar}\hat{g}_\tau(k)\, d\tau.$$

For a thin shell of speed v, with $s = -iE/\hbar$,

$$m_\alpha = \frac{(s - ikv)^{1-\alpha} - (s + ikv)^{1-\alpha}}{2ikv(\alpha - 1)}.$$

Results, under the imposed cone.

(1) On $E > \hbar kc$ the symbol's phase cancels for $\sigma = +1$, so the cone forces the opposite orientation to the Galilean edge. The measured residual is $|\mathrm{Im}|/|\cdot| \le 7 \times 10^{-15}$.
(2) As $\alpha \to 1^-$, $E(k) = \hbar kv \coth(\hbar kv/E_0)$, with $m = 3E_0/(2v^2)$ and $0 < v_g < v$. For a general self-similar profile, $m = 3E_0/(2v_{\text{eff}}^2)$.
(3) Consequently $E_0 \le \frac{2}{3}mc^2$.
(4) For $\alpha < 1$: $E(0) = E_c(E_0/\alpha E_c)^{1/\alpha}$, $m_\alpha = 3E(0)/((1+\alpha)v^2)$, and a cutoff $\hbar kv \approx \frac{1}{2}E_c\big(E_0/(\alpha\varepsilon E_c)\big)^{1/\alpha}$ that recedes as $\varepsilon \to 0$.

The derivations are those of the plane-wave analysis above, applied to m_α. They are kept here, with the numbers below, only as the record of an $\alpha = 2$ ingredient.

k	root	$kv\coth(kv/E_0)$	$E_0 + k^2/2m$	rel. kinetic dev.	v_g/v
0.01	1.000033	1.000033	1.000033	-6.7×10^{-6}	0.007
0.10	1.003331	1.003331	1.003333	-6.7×10^{-4}	0.067
1.00	1.313035	1.313035	1.333333	-6.1×10^{-2}	0.589
3.00	3.014909	3.014909	4.000000	-0.33	0.975

TABLE 10. Imposed cone, thin shell, $v = E_0 = 1$. Not the $\alpha \to 1^-$ edge.

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